

Structural analysis of bioactive glasses

I. Elgayar^{a,*}, A.E. Aliev^b, A.R. Boccaccini^b, R.G. Hill^a

^a Department of Materials, Imperial College London, Exhibition Road, London SW7 2AZ, UK

^b Department of Chemistry, University College London, London WC1H 0AJ, UK

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Abstract

In this study four series of single and mixed alkali glass systems were made and investigated using MAS NMR. Additionally the densities of the glasses were measured experimentally, as well as calculated theoretically using Doweidar's model. MAS NMR was used to obtain a quantitative structural understanding of glasses by calculating the concentrations of bridging and non-bridging oxygens per silicon oxygen tetrahedron as a function of the alkali oxide concentration expressed as Q^n . ^{29}Si MAS NMR spectra exhibited a single resonance corresponding closely with Si in a Q^2 state. The chemical shift of the ^{31}P MAS NMR peak was attributed to phosphate in an orthophosphate environment. The ^{29}Si NMR spectra are in agreement with the density data. Using Doweidar's model the proportions of Q^2 and Q^3 were calculated, showing that all glasses studied are predominantly Q^2 in structure, i.e. $[\text{SiO}_3^-]_n$ silica chains which readily dissolve. The changes in the chemical shifts of the Q^2 and Q^3 species with composition have been interpreted as resulting from the preferential association of Na^+ with Q^3 and Ca^{2+} with Q^2 .

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1. Introduction

Many bioactive glass compositions have good compatibility and high bioactivity. However their application is limited due to their low strength and chemical stability as well as their restricted ease of obtaining the required properties. To be able to design compositions with required properties would be a turning point in bioactive glass formation.

Hench et al. [1], developed the first bioactive glass, Bioglass®, containing 45% SiO_2 , 24.5% Na_2O , 24.4% CaO and 6% P_2O_5 , (all in weight percent) which formed a strong interfacial bond to bone. Its bioactivity is attributed to the formation of a hydroxycarbonate apatite layer on the surface. The rate of tissue bonding appears to depend on the rate of hydroxycarbonate apatite (HCA) formation, which is thought to be formed

by a sequence of reactions between the implanted materials and the surrounding tissues. Three main processes occur: ion exchange, dissolution and precipitation. Ion exchange occurs at the bioactive glass surface, in which cations such as Na^+ and Ca^{2+} from the glass exchange with H^+ or H_3O^+ from the surrounding solution. Network dissolution occurs by the breaking of Si–O–Si bonds by the action of hydroxyl ions (OH^-). The break down of the network results in the release of soluble silica into the solution in the form of silicic acid $[\text{Si}(\text{OH})_4]$. The hydrated silica (Si–OH) formed on the glass surface undergoes rearrangement by polycondensation of neighboring silanols, resulting in a silica rich gel layer. Precipitation of the calcium and phosphate ions released from the glass together with those from solution form a calcium-phosphate rich layer (CaP) on the surface. This mechanism proposed by Hench et al. [1] has been accepted as being the mode of degradation of bioactive glasses. Although it is possible to control the reactivity of the glass by altering its composition, it is difficult to control the degradability of the glass by this route [2].

* Corresponding author.

E-mail address: iman.elgayar@ic.ac.uk (I. Elgayar).

To be able to control the *in vitro* and *in vivo* bioactivity and degradability of melt derived glasses would be of great advantage. The understanding of the glass structure can give information on many of its properties; however there are still many conflicting views on the structure of bioactive glasses [3–8].

Glass structure is very complex, there is no long-range order, but it exhibits some short-to-medium range order. The backbone of the glass silicate structure is the SiO_4 tetrahedral network. These SiO_4 tetrahedra are connected at the corners to form a continuous 3D network. Alkali and alkaline earth oxides serve to modify this network structure, they reduce the degree of connectivity in the network by replacing bridging oxygens (BO) by non-bridging oxygens (NBO). As suggested already by Zachariasen [9] and Warren and Bischoff [10], the network formers form a type of network, with bridging and non-bridging oxygens, while the mobile ions are in the free spaces in between. The network modifiers are important components, since they open up the glass structure and lower the glass transition temperature. The alkali metal ions involved are more or less mobile and can diffuse through the glassy network. Angell [11,12] regards the mobile cations as being effectively decoupled from the glass network.

Based on Raman measurements and model studies it has been suggested by West and Hench [13] that ring systems may exist in the structure of silica glass. However there is no method of direct determination of geometry and structural arrangement in an amorphous system. Previous work done by Rogers and Maries [14] on the formation of crystalline silicate fibres indirectly shows that chain silicates are present within the glass structure. A $\text{CaO} \cdot \text{SiO}_2$ glass was used to produce crystalline rods of wollastonite. The crystallization kinetics of various compositions was determined [14]. The Laue X-ray results showed a very high degree of orientation along the crystallographic *b*-axis, which corresponds to the chain direction of the SiO_4 tetrahedra aligned longitudinally.

Understanding the structure will have a very important part to play in refining the properties of bioactive glasses. Many of the physical properties, such as the transition temperature, softening point and solubility are intimately related to the alkali metal content and to their contribution in the structure. Thus a better knowledge of the structural role of the alkali metal component in particular will lead to an improved understanding of the relationship between structure and properties. The incorporation of a second network modifying alkali metal in a binary alkali metal oxide glass is known to result in a marked deviation in its dynamic or transport properties such as ionic conductivity, diffusivity of network modifying cations and viscosity. This phenomenon known as ‘mixed alkali effect’ has been

observed in many silicate, borate, phosphate, germanate and halide glasses [15].

In addition, mixed alkali oxide compositions enable glass formation at reduced temperatures. The mixed alkali effect can therefore be used to produce glasses with a superior combination of properties than can be obtained with the incorporation of a single alkali metal oxide. To be able to incorporate this mixed alkali effect in bioactive glass compositions would be of great interest. A problem that is normally encountered in bioactive glasses is its high solubility. Mixing alkali metal cations in bioactive glasses could be used to tailor the physical and biological properties.

A number of theories have been proposed over the last forty years in order to explain the mixed alkali effect [16–20]. Many of these theories implicitly or explicitly assume a difference in the bonding and co-ordination environment of the two alkali ions, which results in a hopping site preference of the alkali cations in glass structure [21]. In other words distinct sites are present in the glass structure corresponding to different alkali metal cations, and hopping of one ion to its corresponding site is energetically much more favored than to a site corresponding to the second ion.

The effect of mixing alkali metals in a bioactive glass system has not been investigated in spite of the fact that quite a number of bioactive glasses of mixed alkali compositions have been produced [22]. However they have the potential to be used in incorporating specific characteristics required for certain applications. This study looks at the mixed alkali effect with regards to its structural importance.

Studying the density and NMR spectra of the glasses allows an insight on the structure of the glasses. NMR spectroscopy has played an important role in the development of structural and dynamical models of glass structure. The chemical shift is a sensitive probe of the local environment of a given isotope, hence providing an inventory of the species present. The chemical shift of an individual nucleus is dependent on the shielding of the nucleus (the local electron density which in turn reflects the distribution of other atoms about the nucleus) [23]. It will thus determine the local structural environments present and their compositional dependencies, whereas density data gives an indication of the compactness of the structure.

A more quantitative model for glasses can be obtained by calculating the concentrations of bridging and non-bridging oxygens per silicon oxygen tetrahedron as a function of the alkali oxide concentration. The Q^n notation expresses the concentration of bridging oxygens per tetrahedron, where the value of *n* is equal to the number of bridging oxygen [24]. A tetrahedron fully linked into the network via four bridging oxygens is denoted as a Q^4 unit, while an isolated tetrahedron with no bridging oxygens is denoted as a Q^0 unit.

Density data can also be used to gain insight into the structure of glass. Doweidar and Oraby [25] presented a model to analyze quantitatively the density data of many glass systems. This is done by use of available structural information of the glass to calculate the volume of the structural units. The volume of the structural units in alkali silicate glasses does not change with composition and they depend only on the type of alkali ion, and not the concentration in which they are present [26]. This enables the use of the given model to determine the densities of alkali and mixed alkali silicate glasses with a satisfactory degree of accuracy [27].

Previous work of Lockyer et al. [28], studied similar but slightly different compositions to those studied here, having a higher content of P_2O_5 . Results of the above study showed that the ^{31}P NMR spectra corresponded to phosphate in an orthophosphate environment, indicating that phosphorus is associated with removing sodium and calcium ions from a network modifying role in the silicate network. All the glasses made by Lockyer et al. were annealed for 6 h at $450^\circ C$ before slow cooling. This was originally thought to explain the presence of the orthophosphate as a result of amorphous phase separation. In addition all compositions were made in weight percent, which masks the compositional-property relationships of the bioactive glass. Molar substitutions, therefore have more significance on a structural level, and have been used for this study.

This paper presents ^{29}Si , and ^{31}P NMR results as well as density data of a series of glass compositions. Density measurements will be compared to those calculated using Doweidar's model.

2. Experimental procedures

2.1. Sample preparation

Four series of glass compositions were prepared with varying network connectivities: Na glasses, K glasses, a substitution series of Na for Ca and a mixed alkali series all listed in Table 1. Each sample was prepared from reagent grade $CaCO_3$, Na_2CO_3 , K_2CO_3 , P_2O_5 and SiO_2 .

The compositions were melted in platinum crucibles at different temperatures depending on their transition temperatures for 1 h. The melts were rapidly quenched in water to prevent phase separation. The glass frit was ground in a mill for 14 min and sieved for 30 min in a mechanical shaker to obtain $<45 \mu m$ particles.

Rods of 1 and 2 cm diameters for each composition were cast, by re-melting 50 g of glass frit for a further hour at higher temperatures than the original melting temperature to ensure no bubbles are formed when casting and poured directly into a preheated graphite mould. The glass was allowed to cool slowly to room temperature in the furnace overnight.

2.2. Density measurements

Densities of glass samples were measured using Archimedes' principle with distilled water as the immersion fluid. Rods of 2 cm diameters were used, having been sectioned in such a way to avoid any bubbles that had formed during casting. The measured density was then compared to density calculations using Doweidar's [26] model.

Table 1

The compositions of all the glasses studied (mol%), with their network connectivities (NC) and melting, casting temperatures used

	Glass	SiO_2	P_2O_5	CaO	Na_2O	K_2O	NC	Melting temperature ($^\circ C$)	Casting temperature ($^\circ C$)
Series A	ICIE1	49.46	1.07	23.08	26.38	–	2.04	1370	530
	ICIE2	49.46	1.07	29.64	19.79	–	2.04	1420	570
	ICIE3	49.46	1.07	36.27	13.19	–	2.04	1450	660
	ICIE4	49.46	1.07	42.87	6.6	–	2.04	1475	700
	ICIE5	49.46	1.07	49.46	0.0	–	2.04	1500	780
Series B	ICIE6	48.46	1.07	37.27	13.19	–	1.96	1370	530
	ICIE7	53.46	1.07	33.27	13.19	–	2.29	1450	570
	ICIE8	56.46	1.07	30.27	13.19	–	2.47	1475	660
	ICIE9	61.00	1.07	25.27	13.19	–	2.74	1400	550
Series C	ICIE10	48.46	1.07	37.27	–	13.19	1.96	1370	570
	ICIE11	49.46	1.07	36.27	–	13.19	2.04	1420	650
	ICIE12	53.46	1.07	33.27	–	13.19	2.29	1450	720
	ICIE13	56.46	1.07	30.27	–	13.19	2.47	1475	750
	ICIE14	61.0	1.07	25.17	–	13.19	2.74	1400	550
Series D	ICIE15	48.46	1.07	37.27	6.60	6.60	1.96	1370	570
	ICIE16	49.46	1.07	36.27	6.60	6.60	2.04	1420	620
	ICIE17	53.46	1.07	33.27	6.60	6.60	2.29	1450	670
	ICIE18	56.46	1.07	30.27	6.60	6.60	2.47	1475	750
	ICIE19	61.0	1.07	25.27	6.60	6.60	2.74	1400	550

Table 2

 Q^n species within the different compositional regions [29]

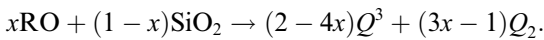
Composition region	Q^n unit formation
Up to 33.3 mol% R_2O/RO	Q^4 units convert into Q^3 units
$33.3 < R_2O/RO < 50$ mol%	Q^3 units convert into Q^2 units
$50.0 < R_2O/RO < 60.0$	Q^2 units convert into Q^1 units
$60.0 < R_2O/RO < 66.7$	Q^1 units convert into Q^0 units

2.3. Calculation of density using Doweidars Model

It has been identified that addition of alkali oxide R_2O/RO create SiO_4 tetrahedra with non-bridging oxygen ions (NBO). Thus the number of NBOs per SiO_4 tetrahedron depends on the alkali oxide concentration [26]. From this, the Q^n species can be calculated. Table 2, defines the compositional regions and type of units within the specified compositional ranges.

Up to 33.3 mol% R_2O , Q^4 units convert into Q^3 units, where Q^4 denotes to SiO_4 tetrahedral having four bridging oxygen ions and Q^3 units are SiO_4 tetrahedra having three bridging oxygen ions and one NBO ion. At 33.3 mol% R_2O the glass is composed of Q^3 units only. On increasing the alkali oxide content further, Q^3 units convert progressively into Q_2 units; the latter contain two NBO's per unit. This process goes on until complete conversion is achieved at 50 mol% R_2O . At this composition there are only Q_2 units in the glass [29].

Doweidar [26] uses these conversions of the Q^n units to calculate the density of the glass within its compositional range. In the region $33.3 < x < 50$ mol% relevant to bioactive glasses, the conversion of Q^3 units into Q_2 can be given as:



Within this region the density can be given as:

$$D(x) = [(2 - 4x)M_3 + (3x - 1)M_2] / [(2 - 4x)V_3 + (3x - 1)V_2], \quad (1)$$

where M_2 is the mass of Q_2 units that contains one R^{2+} ion (the mass of $Si + 3O + R$) and V_2 is its volume.

The glasses studied in series A (effect of substituting Na for Ca), B (Na, Ca-glasses with differing network connectivities) and C (K, Ca-glasses with differing network connectivities), are in the range of $33.3 < [R_2O]_a + [RO]_b < 50$ mol% with molar composition of $x_a(R_2O) \cdot x_b(R_2O)_b \cdot (1 - x_a - x_b)SiO_2$, where the modifier oxides are $(R_2O)_a$ and $(R_2O)_b$, and the mol fractions are x_a and x_b . Within this range some of the Q^3 units convert into Q_2 units. The number of Q^{3a} , Q^{3b} , Q^{2a} , and Q^{2b} depends on the value of x_a and x_b and were calculated from the glass molar composition, where $x_t = (x_a + x_b)$.

$$\begin{aligned} & x_a(R_2O)_a + x_b(R_2O)_b + (1 - x_a - x_b)SiO_2 \\ & \rightarrow (2 - 4x_t)(x_a/x_t)Q^{3a} + (2 - 4x_t)(x_b/x_t)Q^{3b} \\ & + (3x_t - 1)(x_a/x_t)Q^{2a} + (3x_t - 1)(x_b/x_t)Q^{2b}. \end{aligned}$$

Table 3

Structural unit volumes of the silicate units in glasses studied [26]

Type of glass	Modifier ion	V_3 (10^{-24} cm^3)	V_2 (10^{-24} cm^3)
$Na_2O-CaO-SiO_2$	Na	60.5	79
	Ca	54.5	66.4
$K_2O-CaO-SiO_2$	K	72	102
	Ca	55.4	67.2
$Na_2O-K_2O-CaO-SiO_2$	Na	60.5	79
	K	72	101.7
	Ca	54.5	66.4

Since the volume of structural units in alkali silicate glasses does not change with composition and depends only on the type of alkali ion present, but not on the concentration, Doweidar incorporated this to determine the density of the glass. Table 2 shows the unit volumes of the components of the glasses studied. Eq. (2) shows the calculation used to determine the densities of the glasses in series A, B and C.

$$\begin{aligned} D = \{ & (2 - 4x_t)(x_a/x_t)M_{3a} + (2 - 4x_t)(x_b/x_t)M_{3b} \\ & + (3x_t - 1)(x_a/x_t)M_{2a} + (3x_t - 1)(x_b/x_t)M_{2a} \} \\ & / \{ (2 - 4x_t)(x_a/x_t)V_{3a} + (2 - 4x_t)(x_b/x_t)V_{3b} \\ & + (3x_t - 1)(x_a/x_t)V_{2a} + (3x_t - 1)(x_b/x_t)V_{2a} \}, \quad (2) \end{aligned}$$

where M_{2a} is the mass of the $Q^2(Na)$ unit and M_{2b} is the mass of the $Q^2(Ca)$ unit. V_{2a} and V_{2b} are respectively the volumes of these units. Table 3 shows the volumes of the silicate units in the glasses studied.

Series D encompassed mixed alkali glasses ($Na_2O-K_2O-CaO-SiO_2-P_2O_5$), containing different modifier cations. In order to determine the density, Eqs. (1) and (2) are modified to include the third modifier oxide. Therefore in the region of $33.3 < (x_a + x_b + x_c) < 50$ mol% the density would be given as shown in Eq. (3) below where $x_t = x_a + x_b + x_c$

$$\begin{aligned} D = \{ & (2 - 4x_t)(x_a/x_t)M_{3a} + (2 - 4x_t)(x_b/x_t)M_{3b} \\ & + (2 - 4x_t)(x_c/x_t)M_{3c} + (3x_t - 1)(x_a/x_t)M_{2a} \\ & + (3x_t - 1)(x_b/x_t)M_{2b} + (3x_t - 1)(x_c/x_t)M_{2c} \} \\ & / \{ (2 - 4x_t)(x_a/x_t)V_{3a} + (2 - 4x_t)(x_b/x_t)V_{3b} \\ & + (2 - 4x_t)(x_c/x_t)V_{3c} + (3x_t - 1)(x_a/x_t)V_{2a} \\ & + (3x_t - 1)(x_b/x_t)V_{2a} + (3x_t - 1)(x_c/x_t)V_{2c} \} \quad (3) \end{aligned}$$

However, Doweidar's model does not account for the phosphate content in the glasses. Since 1.07% P_2O_5 was used in all of the glasses this content was considered negligible when applying Doweidar's model.

2.4. NMR measurements

Solid-state ^{29}Si and ^{31}P NMR spectra were recorded on a Bruker MSL300 spectrometer (7.05 T) at the resonance frequency of 59.63 and 121.49 MHz, respectively, using standard Bruker double-resonance magic angle

sample spinning (MAS) probes. The glass powder was packed into a cylindrical zirconia rotor (7 mm external diameter) and then spun at MAS frequency in the range 4.0–5.5 kHz (with stability better than approximately ± 3 Hz) for ^{29}Si MAS NMR. Solid-state ^{29}Si NMR spectrum of one of the glass samples was also recorded on a Bruker AMX600 spectrometer at the resonance frequency of 119.2 MHz using Bruker 4 mm MAS probe. For ^{31}P MAS NMR a 4 mm outer-diameter zirconia rotors were used and the MAS frequency was in the range 10–11 kHz. A 20° pulse was used for both ^{29}Si and ^{31}P and recycle times used were 60 s for ^{29}Si and 30 s for ^{31}P . Longer delays were explored for both ^{29}Si and ^{31}P NMR measurements, but no changes in signal intensity and lineshape were observed. All spectra were recorded at ambient probe temperature. The ^{29}Si and ^{31}P chemical shifts are given relative to tetramethylsilane and 85% aqueous solution of H_3PO_4 , respectively.

3. Results and discussion

3.1. Density measurements

The results of the density measurements are summarized in Figs. 1 and 2, and show that there is generally a good agreement between the measured densities and those calculated by Doweidar's model.

Fig. 1 shows that as the concentration of alkali oxide is increased, the density decreases, which can be explained by the increase of network disruption with the increase of the Na_2O content, which in turn leads to a less dense structure. Additionally, as it follows from Fig. 2, with increasing network connectivity the densities of the sodium glasses are higher than those of the complementary potassium glasses. Doweidar [25] suggests that this is attributed to the higher density (D) of the

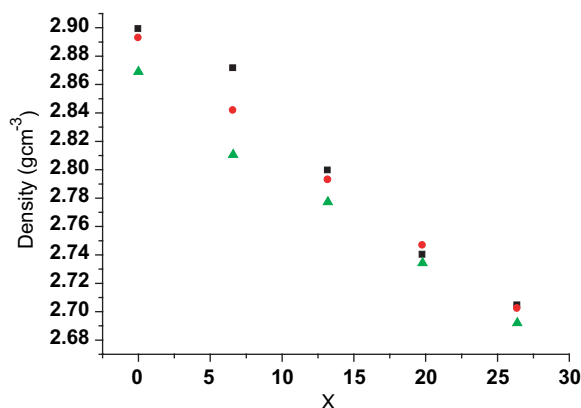


Fig. 1. Change of density with increasing alkali metal concentration in the series, $3\text{SiO}_2 \cdot 0.07\text{P}_2\text{O}_5 \cdot (3 - X)\text{CaO} \cdot X\text{Na}_2\text{O}$, where X is the molar concentration of Na_2 . (▲) shows data taken from Ref. [30], (●) indicates the theoretical values calculated using Doweidar's model, (■) indicates the experimentally measured densities.

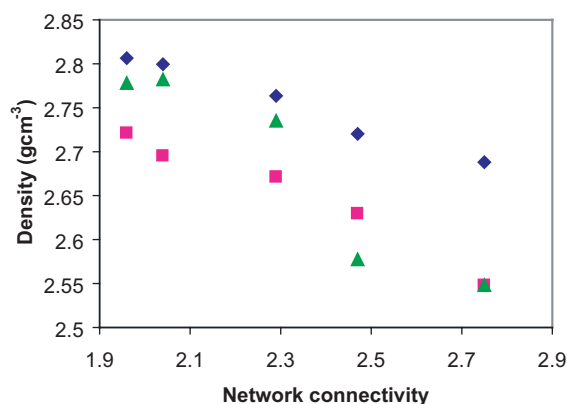


Fig. 2. Experimentally calculated densities of series B, C and D, (◆) Na glasses, (■) K glasses and (▲) mixed alkali glasses respectively.

$Q^2(\text{Na})$ units compared with that of the $Q^2(\text{K})$ units, 2.5667 and 2.5201 g cm^{-3} , respectively. In addition, from the crystal ionic radii for Na^+ (0.97 Å) and K^+ (1.33 Å) [31], it can be estimated that for hypothetical materials consisting of Na^+ and K^+ the corresponding densities are related by: $D(\text{Na}^+) \approx 1.5D(\text{K}^+)$. Hence, a glass containing Na^+ is expected to have higher density than a glass containing K^+ , and this agrees with the measurements shown in Fig. 2. The mixed alkali glasses have densities intermediate between those for the equivalent single alkali containing glasses. Most importantly the measured densities have a good correlation with those calculated using Doweidar's equation as shown in Fig. 3, thus confirming that the glasses are predominantly Q^2 in structure, and all the assumptions made when calculating Q^n species can be accepted. However a significant deviation from theory is observed for the mixed alkali glasses.

Table 4 shows the proportions of Q^2 and Q^3 in the glasses studied, calculated using Doweidar's model and

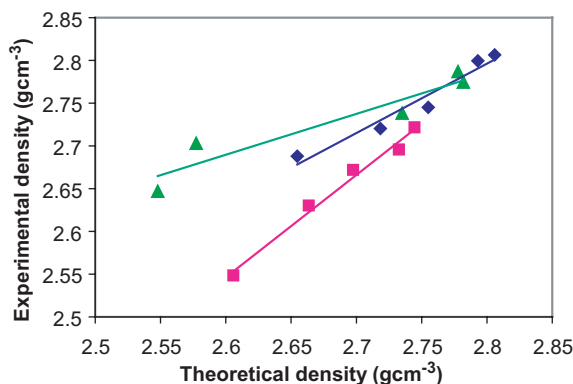
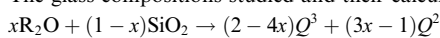


Fig. 3. Comparison of theoretical and experimental density on glass series B, C and D, assuming binary distribution of Q -species (Q^2 and Q^3). (◆) shows the Na glasses, (■) the K glasses, and (▲) presents the mixed alkali glasses respectively.

Table 4

The glass compositions studied and their calculated Q^n proportions

Glass	Composition (in mol%)				Proportion of Q^3	Proportion of Q^2
	SiO ₂	CaO	Na ₂ O	K ₂ O		
ICIE1	0.4946	0.2308	0.2638	–	4.3	95.7
ICIE2	0.4946	0.2968	0.1979	–	4.3	95.7
ICIE3	0.4946	0.3627	0.1319	–	4.3	95.7
ICIE4	0.4946	0.4287	0.066	–	4.3	95.7
ICIE5	0.4946	0.4946	0	–	4.3	95.7
ICIE6	0.4846	0.3727	0.1319	–	2.5	97.5
ICIE3	0.4946	0.3627	0.1319	–	4.3	95.7
ICIE7	0.5346	0.3327	0.1319	–	26.4	73.6
ICIE8	0.560	0.3027	0.1319	–	46.1	53.9
ICIE9	0.61	0.2527	0.1319	–	64.0	36.0
ICIE10	0.4846	0.3727	–	0.1319	2.5	97.5
ICIE11	0.4946	0.3627	–	0.1319	4.3	95.7
ICIE12	0.5346	0.3327	–	0.1319	26.4	73.6
ICIE13	0.560	0.3027	–	0.1319	46.1	53.9
ICIE14	0.61	0.2527	–	0.1319	64.0	36.0
ICIE15	0.4846	0.3727	0.066	0.066	2.5	97.5
ICIE16	0.4946	0.3627	0.066	0.066	4.3	95.7
ICIE17	0.5346	0.3327	0.066	0.066	26.4	73.6
ICIE18	0.560	0.3027	0.066	0.066	46.1	53.9
ICIE19	0.61	0.2527	0.066	0.066	64.0	36.0

disregarding the P₂O₅ content. Table 5 shows similar data, but calculated by taking into account the molar percent of P₂O₅. These proportions of Q^2 and Q^3 calcu-

lated using Doweidar's model were used to explain the observed chemical shift differences in ²⁹Si MAS NMR spectra of the glasses studied.

Table 5

The proportions of Q^2 and Q^3 of glasses studied and those from the work of Lockyer et al. [28] taking into account the molar percent of P₂O₅

Glass	Composition				Proportion of Q^3	Proportion of Q^2
	SiO ₂	CaO	Na ₂ O	K ₂ O		
ICIE1	0.4946	0.2308	0.2638	–	16.45	83.55
ICIE2	0.4946	0.2968	0.1979	–	16.45	83.55
ICIE3	0.4946	0.3627	0.1319	–	16.45	83.55
ICIE4	0.4946	0.4287	0.066	–	16.45	83.55
ICIE5	0.4946	0.4946	0	–	16.45	83.55
ICIE6	0.4846	0.3727	0.1319	–	8.95	91.05
ICIE3	0.4946	0.3627	0.1319	–	16.45	83.55
ICIE7	0.5346	0.3327	0.1319	–	37.32	62.68
ICIE8	0.560	0.3027	0.1319	–	56.03	43.97
ICIE9	0.61	0.2527	0.1319	–	83.27	16.73
ICIE10	0.4846	0.3727	–	0.1319	8.95	91.05
ICIE11	0.4946	0.3627	–	0.1319	16.45	83.55
ICIE12	0.5346	0.3327	–	0.1319	37.32	62.68
ICIE13	0.560	0.3027	–	0.1319	56.03	43.97
ICIE14	0.61	0.2527	–	0.1319	83.27	16.73
ICIE15	0.4846	0.3727	0.066	0.066	8.88	91.12
ICIE16	0.4946	0.3627	0.066	0.066	16.38	83.62
ICIE17	0.5346	0.3327	0.066	0.066	37.25	62.75
ICIE18	0.560	0.3027	0.066	0.066	55.97	44.03
ICIE19	0.61	0.2527	0.066	0.066	83.22	16.78
Comp. from Lockyer et al.	0.489	0.105	0.379	–	39.6	60.4
	0.487	0.156	0.330	–	38.3	61.7

3.2. ^{29}Si NMR

The proportions of different Q species in glasses can be determined using NMR techniques, assuming that the Q species are randomly distributed or clustered [32]. The Q -resolution capacity of ^{29}Si NMR is in general based on the fact that the different Q species have different ^{29}Si chemical shifts due to the differences in their electronic environment, as well as slight deviations in their geometry. However, in the case of silicate glasses broad signals are usually observed, which considerably obscure the analysis of the spectra. Two-dimensional ^{29}Si NMR experiment correlating the MAS spectrum and the off-MAS spectrum has been used for Q^n -quantification [33,34], however, so far the application of this technique has been restricted to 95% ^{29}Si -enriched glasses. In addition, a special DAS probe is required for this type of experiments.

Fig. 4 illustrates isotropic parts of typical ^{29}Si NMR spectra recorded for glasses and consisting of a slightly asymmetric peak with a peak maximum at approximately -80 ppm. The observed linewidths measured at

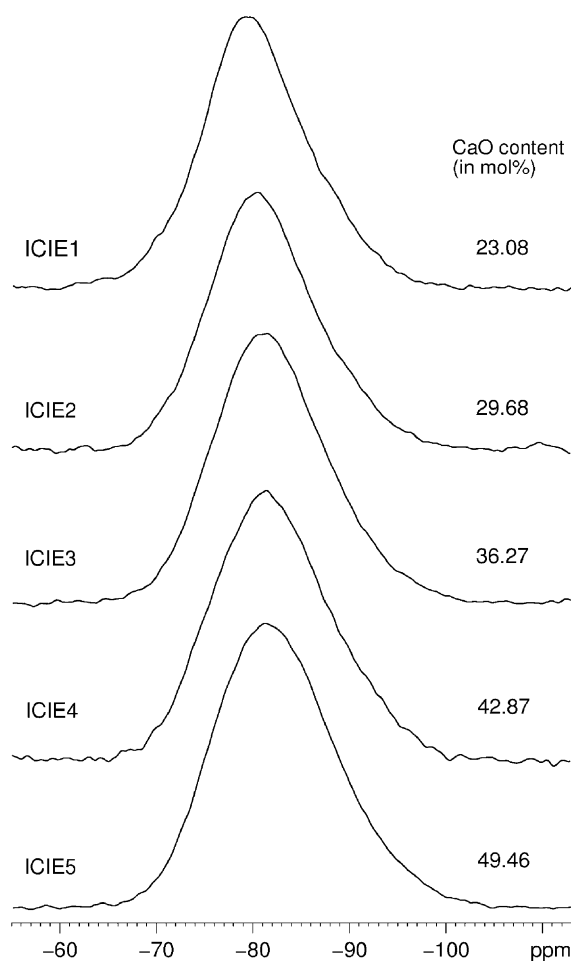


Fig. 4. The ^{29}Si MAS NMR spectra recorded for glasses ICIE1–ICIE5 (series A).

half-height for all the glasses studied were in the range 14 ± 3 ppm, which is larger than the expected chemical shift difference for Q^n and Q^{n+1} species (≈ 5 ppm for Q^2 and Q^3 and ≈ 10 ppm for Q^3 and Q^4). Despite the broad lines observed, the first inspection of the ^{29}Si spectra of the glasses studied nevertheless reveals that the majority of them are predominantly Q^2 in structure, as the peak maxima are in the range -79.6 to -81.7 ppm (Table 6). This region of ^{29}Si chemical shifts is characteristic for Q^2 species [28,33,34].

In comparison, compositions made by Lockyer et al. [28] were similar but slightly different to those in this study, having slightly higher P_2O_5 content (2.6 mol%). However they also studied compositions with no CaO content and with 10% CaO substitution. ^{29}Si NMR spectra for these compositions showed a clear indication of two resonances with a degree of overlap, corresponding to Q^2 - and Q^3 -sites. At higher CaO content (>15.1 mol%), the peaks merge into a single peak with a shoulder on the Q^3 side.

In analogy with the previous studies, for series A in the present work we assume that the observed asymmetric lineshape with a shoulder near the Q^3 peak position is a superposition of two overlapping signals due to Q^2 (at approximately -80 ppm) and Q^3 (at approximately -90 ppm) species with predominant population of the Q^2 sites. The signal overlap is caused by the large linewidths of the Q^2 and Q^3 peaks, which are due to a continuous distribution of isotropic ^{29}Si chemical shifts. The latter itself is a manifestation of a continuous distribution of local structural factors in glasses, such as O–Si–O bond angles and oxygen-metal distances. Overall for the glasses studied in this work the peak maxima are in the range -79 to -90 ppm, which is characteristic for Q^2 and Q^3 species. The predominant Q^2 structure results are in agreement with Doweidar's analysis shown in Table 5.

Further inspection of the observed peak maxima and linewidths shows that these two parameters are dependent on the glass composition within each series (Table 6). As an example, Fig. 5 shows that as Na_2O is substituted for CaO, (series A with the constant SiO_2 content) the position of the peak maximum shifts to a lower frequency in an almost linear fashion with increasing CaO content. Based on the Doweidar's model (Tables 4 and 5) it is assumed that within series A the Q^2/Q^3 ratio is constant, and hence the observed spectral changes are likely to be caused by the change of the cation. Taking into account the chemical shift values reported for Q^2 species in Na_2SiO_3 ($\delta_{\text{Si}} = -76$ ppm) [35] and for CaSiO_3 ($\delta_{\text{Si}} = -81.5$ ppm) [36] the observed shift of the peak maxima for series A can be explained by the increase of the proportion of Q^2 species with non-bridging oxygens neutralized by calcium ions (denoted as $Q^2(\text{Ca})$). Also, as shown in Table 6, the linewidth of the peak increases with increasing CaO content, indicating that

Table 6

The spectral parameters determined from the ^{29}Si and ^{31}P MAS NMR studies of glasses ICIE1–ICIE19

Glass	^{29}Si NMR		^{31}P NMR	
	Peak position (ppm)	Linewidth (ppm)	Peak position (ppm)	Linewidth (ppm)
ICIE1	−79.6	11.2	9.5	7.8
ICIE2	−80.2	11.6	7.3	7.8
ICIE3	−81.2	12.7	5.3	7.7
ICIE4	−81.4	13.3	3.7	7.7
ICIE5	−81.7	13.8	2.9	8.0
ICIE6	−80.4	12.7	5.4	7.6
ICIE3	−81.3	12.7	5.6	7.6
ICIE7	−80.9	12.6	5.5	7.5
ICIE8	−84.4	15.4	5.5	8.1
ICIE9	−88.9	15.8	5.5	8.9
ICIE10	−81.0	13.6	4.9	7.5
ICIE11	−81.6	12.6	4.8	7.5
ICIE12	−84.0	15.1	4.5	7.9
ICIE13	−84.9	14.4	4.7	8.1
ICIE14	−89.9	16.1	4.6	9.4
ICIE15	−81.1	12.6	5.4	7.4
ICIE16	−81.5	11.4	5.6	7.2
ICIE17	−83.5	15.6	5.0	8.0
ICIE18	−84.6	16.3	5.0	6.0
ICIE19	−89.4	16.2	5.1	8.6

$Q^2(\text{Ca})$ and $Q^3(\text{Ca})$ species are structurally disordered to a larger extent when compared to corresponding $Q^2(\text{Na})$ and $Q^3(\text{Na})$.

We note that unlike the early work by Lockyer et al., which has used mainly a binary distribution of Q^2 and Q^3 to describe ^{29}Si spectra (similar to those in Fig. 4), subsequent reports have made use of more complex models based on the disproportionation reaction:

$$Q^n = Q^{n-1} + Q^{n+1}.$$

In the case of ^{29}Si -enriched CaSiO_3 , for example, relative amounts of all five Q -species have been reported. Although fittings of the large number of cross-sections of the two-dimensional spectrum were employed, a quality of fit for simple two-site or three-site distributions was not assessed. Nevertheless, based on the relative amounts for five-site distribution model (0.7% for Q^0 and 1.1% for Q^4 [33]), it is clear that for glasses with single ^{29}Si peak maxima at approximately −80 ppm, the model can be restricted to a three-site distribution of Q^1 , Q^2 and Q^3 . Occurrence of Q^0 and Q^4 species (with corresponding ^{29}Si chemical shifts at −70 ppm and −110 ppm, respectively) for the glasses of the composition similar to that of series A is rare [28]. Inspection of the ^{29}Si spectra shows that for all the glasses of series A there are no peaks or discontinuities at approximately −105 ppm or −70 ppm, hence no Q^4 or Q^0 species are present in these glasses.

Distinction of Q^1 and Q^2 is most problematic. Chemical shift for Q^1 species is expected to be approximately −75 ppm [33], which is close to that of Q^2 species (approximately −80 ppm). Further complication arises

from the fact that in mixed metal glasses a distribution of ^{29}Si chemical shifts is expected for each Q^n , for example, $\delta_{\text{Si}} = -76$ ppm for $Q^2(\text{Na})$ [35] and $\delta_{\text{Si}} = -81.5$ ppm for $Q^2(\text{Ca})$ [36]. Strictly speaking identification of Q^1 species (minor compared to Q^2) based on ^{29}Si NMR data alone is difficult. Therefore, in the absence of adequate spectral resolution, we have adopted a simple binary distribution model, similar to that used by Lockyer et al. [28] and restricted mainly to Q^2 and Q^3 species, which is sufficient to describe the observed changes in ^{29}Si spectra depending on the composition of the glasses studied.

In general, for the type of lineshapes observed the use of deconvolution techniques for extracting peak intensities from overlapping spectra may not lead to a unique solution. Hence, additional constraints are needed in order to verify the found solution. For one of the glasses (ICIE1) an additional ^{29}Si spectrum was measured at 11.1 T. This glass has the lowest CaO content, hence it is expected to have the smallest linewidths for the contributing Q^2 - and Q^3 -species. The availability of two ^{29}Si spectra at two different fields was used as a constraint for the verification of the deconvolution results. Independent fittings of these two spectra showed that the use of the binary distribution model is sufficient and the two solutions found within this model are similar (Table 7). In addition, the optimized chemical shift values are close to those expected for Q^2 and Q^3 species.

For series B, C and D a more pronounced chemical shift changes from approximately −80 to −90 ppm and linewidths changes from approximately 13 to 16 ppm are observed as the SiO_2 content is gradually increased

Table 7

Chemical shifts, linewidths and percent contribution of Q^2 (approximately -80 ppm) and Q^3 (approximately -89 ppm) species in ^{29}Si spectra of ICIE1 measured at two different fields

Field	δ_{Si} (ppm)	Linewidth (ppm)	Contribution (%)
7.05 T	79.4	11.3	89
	-89.1	8.2	11
11.1 T	-79.4	11.3	88
	-89.4	8.7	12

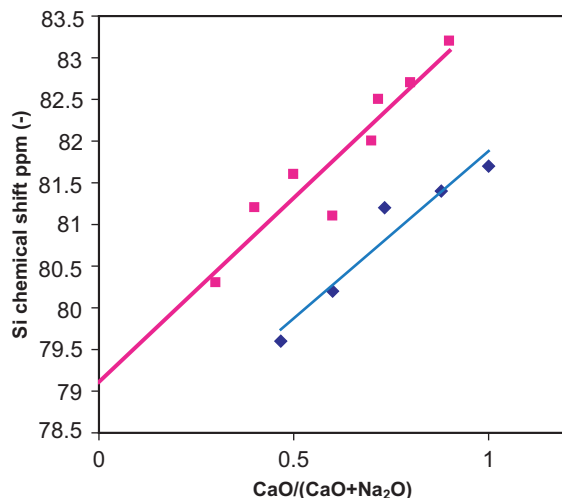


Fig. 5. Comparison of Si chemical shift of series A (◆), with similar compositions made by Lockyer et al. [28] (■).

from 48.5 mol% to 61 mol% (Table 6). These spectral changes clearly indicate that the content of Q^3 species is increasing within each series. Based on the observed chemical shifts, it is concluded that glasses containing 61 mol% SiO_2 [ICIE9 (series B), ICIE14 (series C) and ICIE19 (series D)] with peak maxima at -89 to -90 ppm are predominantly Q^3 in structure, whereas glasses containing 48.5 mol% SiO_2 [ICIE6 (series B), ICIE10 (series C) and ICIE15 (series D)] with peak maxima at -80 to -81 ppm are predominantly Q^2 in structure.

Fig. 6 presents the ^{29}Si NMR chemical shift plotted against the calculated Q^3 content for the compositions of series B, C, and D with differing network connectivities. It is seen that the Na glasses, K glasses and the mixed alkali series all show the same trend with the decreased chemical shift increasing with Q^3 content. There is an approximately linear correlation. In addition, the data presents a negative chemical shift with increase in Si content. This indicates that an increase in Si content in the glass causes an increase in network connectivity, thus resulting in a reduction of the number of non-bridging oxygens within the glass and an increase of the proportion of Q^3 .

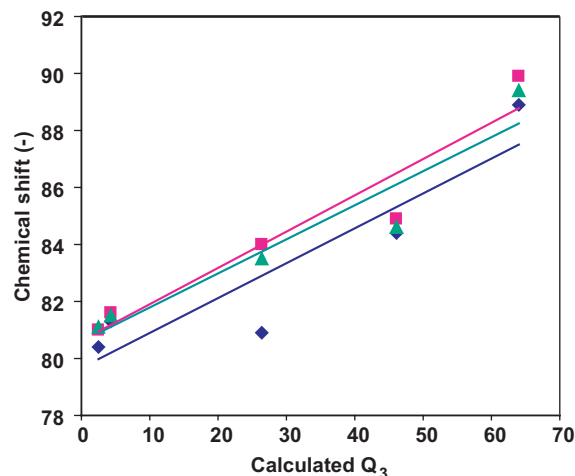


Fig. 6. Si chemical shift for the K glasses (■), Na glasses (◆) and the mixed alkali glass series (▲).

3.3. ^{31}P NMR

^{31}P NMR spectra for series A glasses show a single resonance with approximately constant linewidth (≈ 7.8 ppm) and with the chemical shift in the range 2.9–9.5 ppm (Table 6). Previously PO_4^{3-} was assumed to enter the glass network as a network former as shown in Fig. 7(a).

However ^{31}P NMR spectra correspond to an orthophosphate environment, indicating that phosphorus is associated with removing the sodium and calcium ions from a network-modifying role in the silicate network, see Fig. 7(b). The work of Lockyer et al. [28] shows agreement with these findings; results obtained from their study corresponded to a single resonance assigned to an orthophosphate-like environment. All the glasses made by Lockyer et al. were annealed for 6 h at 450°C before slow cooling. This was originally thought to explain the presence of the orthophosphate as a result of amorphous phase separation. However, the compositions made in the current study were rapidly quenched in water, reducing possibilities of phase separation.

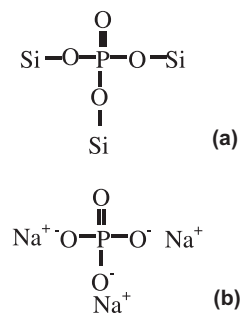


Fig. 7. (a) The role of phosphate as a network former. (b) An orthophosphate structure.

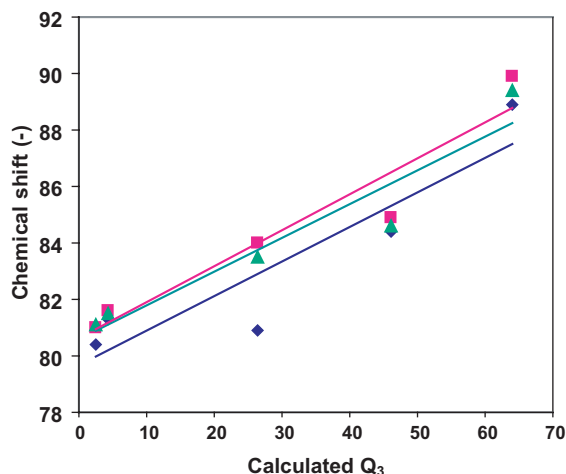


Fig. 8. Comparison of ^{31}P chemical shifts of series A, and similar compositions studied by Lockyer et al. [28].

As it follows from Table 6, replacing Na_2O with CaO in series A glasses results in shifts to lower frequencies in ^{31}P spectra, a trend that continues as the amount of CaO increases, as seen in Fig. 8. The orthophosphate shift can be accounted for by the electronegativities between Na^+ and Ca^{2+} , (0.9 and 1.0 respectively) [28]. Additionally, the fact that sodium and calcium charge balance the orthophosphate groups will result in a reduction in the number of non-bonding oxygens attached to Si and an increase in Q^3 content.

In the glasses studied the low molar percent of P_2O_5 used in the compositions is small and consequently only a small portion of Q^3 will be present. The glasses made by Lockyer et al. show a higher proportion of Q^3 largely as a result of containing a higher percentage of P_2O_5 . Table 5 shows the Q^3 and Q^2 proportions of the compositions studied, calculated by using Doweidar's model and by taking into account the molar percent of P_2O_5 . These calculations also assume that the phosphate is present as orthophosphate. It can be seen that the proportion of Q^3 in glasses made by Lockyer et al. are considerably higher than the compositions investigated in this study. The increase in the proportion of Q^3 species is a result of Na^+ and Ca^{2+} ions locally charge balancing orthophosphate species rather than non-bridging oxygens attached to silicon.

For glasses of series B, C and D, to the first approximation ^{31}P chemical shifts are independent of the composition within each series, whereas the linewidth increases as CaO is substituted by SiO_2 (Table 6). Closer inspection of spectra reveals that the linewidth increase is caused by the appearance of a shoulder at lower frequency with a peak maximum at approximately -9 ppm relative to the major peak (Fig. 9). The intensity of the low-frequency component reaches maximum for the glasses with the highest SiO_2 content studied, i.e. ICIE9, ICIE14 (Fig. 9) and ICIE19. Deconvolution of

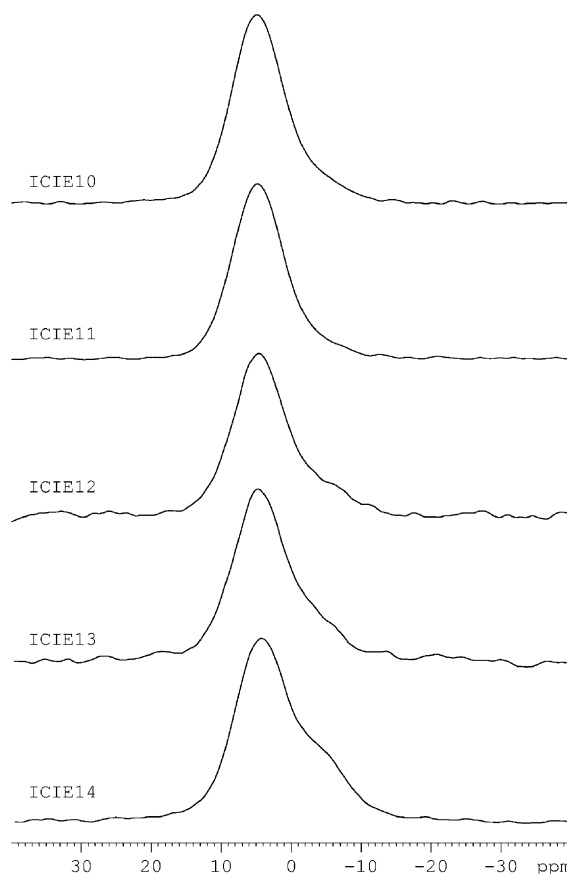


Fig. 9. The ^{31}P MAS NMR spectra recorded for glasses ICIE10–ICIE14 (series C).

Table 8
Chemical shifts, linewidths and percent contribution of two components in ^{31}P spectra of glasses ICIE9, ICIE14 and ICIE19

Glass	δ_P (ppm)	Linewidth (ppm)	Contribution (%)
ICIE9	5.8	8.2	75
	−3.0	9.8	25
ICIE14	4.6	8.3	74
	−4.4	8.9	26
ICIE19	5.2	8.0	76
	−3.6	9.4	24

the observed lineshapes for these glasses were carried out and the results are summarized in Table 8.

In line with the previous studies [28,37,38], the major component ($\approx 75\%$) is assigned to the orthophosphate species, PO_4^{3-} , whereas the minor component ($\approx 25\%$) is assigned to the diphosphate species, $\text{P}_2\text{O}_7^{3-}$. As can be seen from Table 8, the chemical shifts for $\text{Na}_2\text{O}+\text{K}_2\text{O}$ glass ICIE19 for both PO_4^{3-} and $\text{P}_2\text{O}_7^{3-}$ are intermediate between those observed for Na_2O glass ICIE9 and K_2O glass ICIE4. Hence, it is likely that the observed small chemical shift differences between series B, C and D glasses are caused by the small differences in the electronegativities of the alkaline cations.

4. Conclusion

An increase in Si content in the glass causes an increase in network connectivity, thus resulting in a reduction of the number of non-bridging oxygens and an increase of the proportion of Q^2 to Q^3 . These two Q species have been observed co-existing for compositions in other studies. However, single resonances are seen in the ^{29}Si MAS NMR in this research, although it is assumed that two separate resonances are present which cannot be resolved. The NMR data from our study and that of Lockyer et al. [28] show very similar results even though all the glasses made by Lockyer et al. were annealed for 6 h at 460 °C before cooling slowly, while the glasses in this study were melted at different temperatures depending on their T_g for 1 h then quenched in water. Both results indicate that P is present in an orthophosphate form rather than taking part in the silicate network. The ^{31}P NMR spectrum indicates that there is an association between Na^+ and orthophosphate (PO_4^{3-}).

Both the density and NMR results indicate that the glasses studied are predominantly Q^2 in structure. The NMR data shows little difference between the single alkali compositions and the mixed alkali glasses. However the density data for the mixed alkali glasses shows a marked deviation from that predicted by Doweidar's model. This suggests some type of structural change occurs for the mixed alkali glasses. To further analyze the mixed alkali compositions, dissolution studies need to be carried out.

Understanding the structure will have a very important part to play in refining the properties of many bioactive glasses as many of the physical properties are intimately related to the alkali metal content and to their contribution in the structure. Thus a better knowledge of the structural role of the alkali metal component and the Q^n proportions required will lead to an improved understanding of the relationship between structure and properties.

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